

41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Week 2: Cloud Infrastructure Design: Virtualization Technologies

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- * Assessment 1 is released.

Assignment 1 - DUE DATE: Week 5 (17:00 **Friday 28 August 2026**, weight: 20%)

- * Late submissions:

- * A late penalty of 20% per working day will apply.
- * Special consideration for late submission must be arranged by Week 4. Formal documentation must be provided.

Contents

- * Definitions and examples of cloud service models
- * Infrastructure components
 - * Hardware: Computing, Storage, Network
 - * Software: Virtualization Software, VMM
- * Virtualization
 - * Virtual Machines and Hypervisors
 - * Storage Virtualization
 - * Network Virtualization

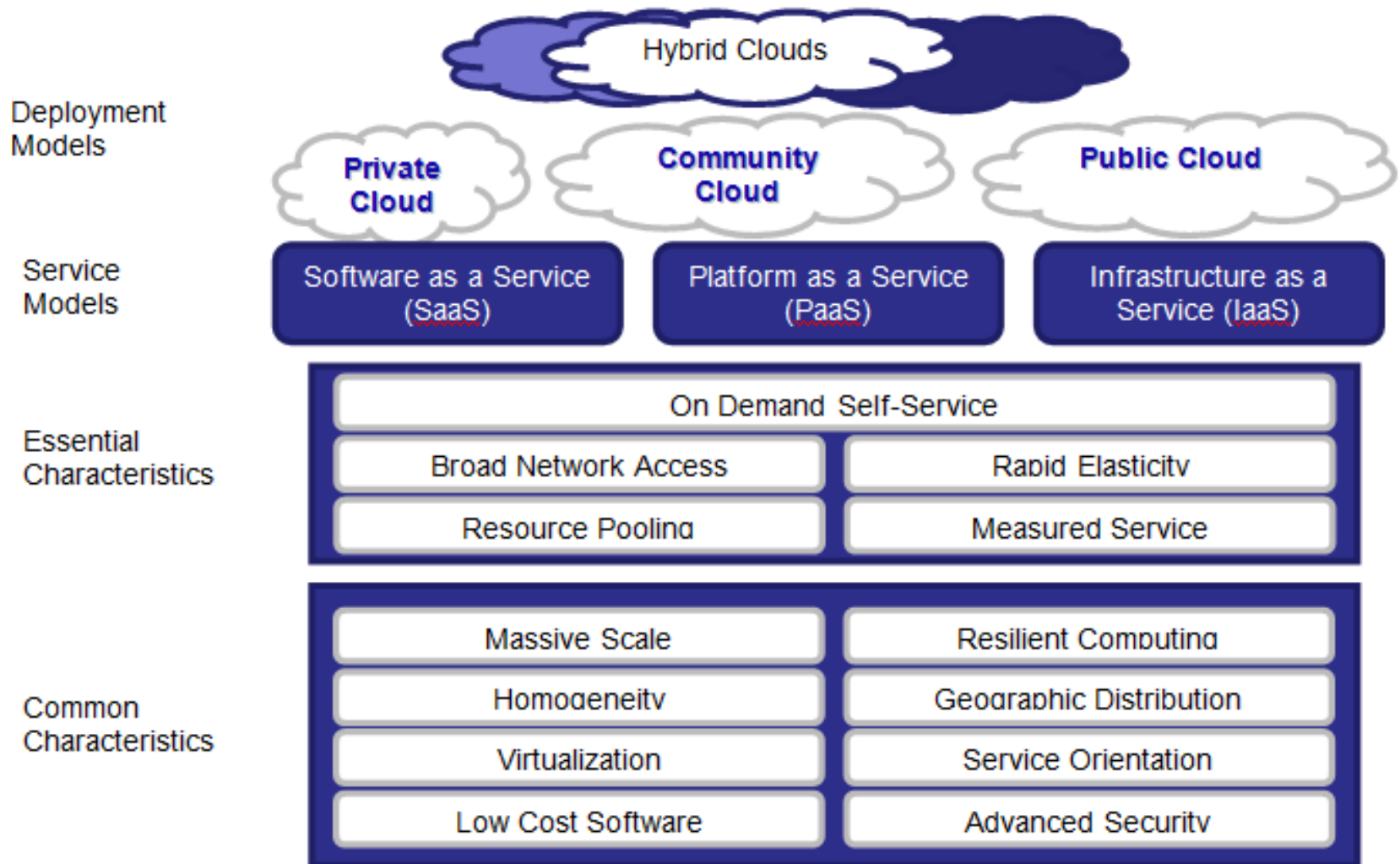
Definitions

- * National Institute of Standards and Technology (NIST):
*“Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction”*¹
- * Thomas Erl:
*“Cloud computing is a specialized form of distributed computing that introduces utilization models for remotely provisioning scalable and measurable resources”*²

1. <https://www.nist.gov/news-events/news/2011/10/final-version-nist-cloud-computing-definition-published>

2. **Cloud Computing:** Concepts, Technology & Architecture, book by Thomas Erl, Ricardo Puttini, Zaigham Mahmood

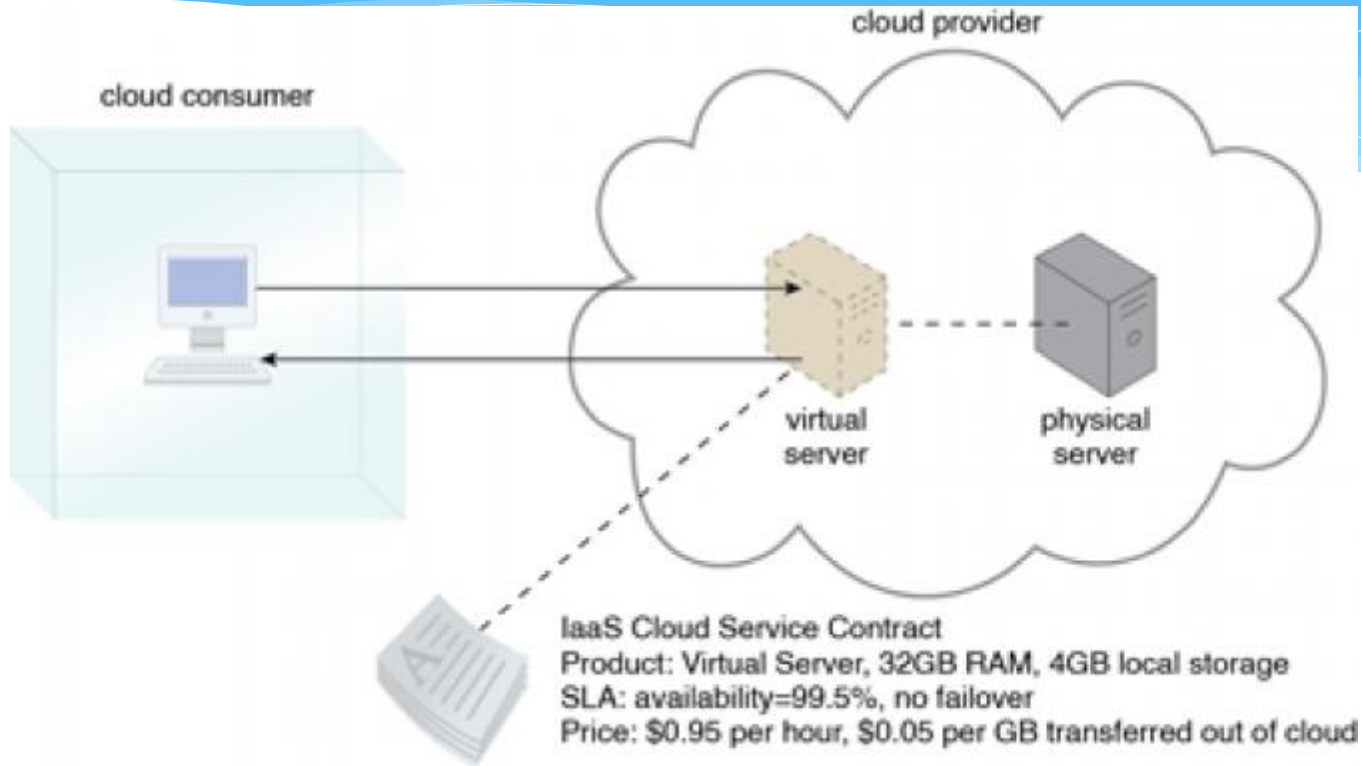
NIST Cloud Definition



Terminology

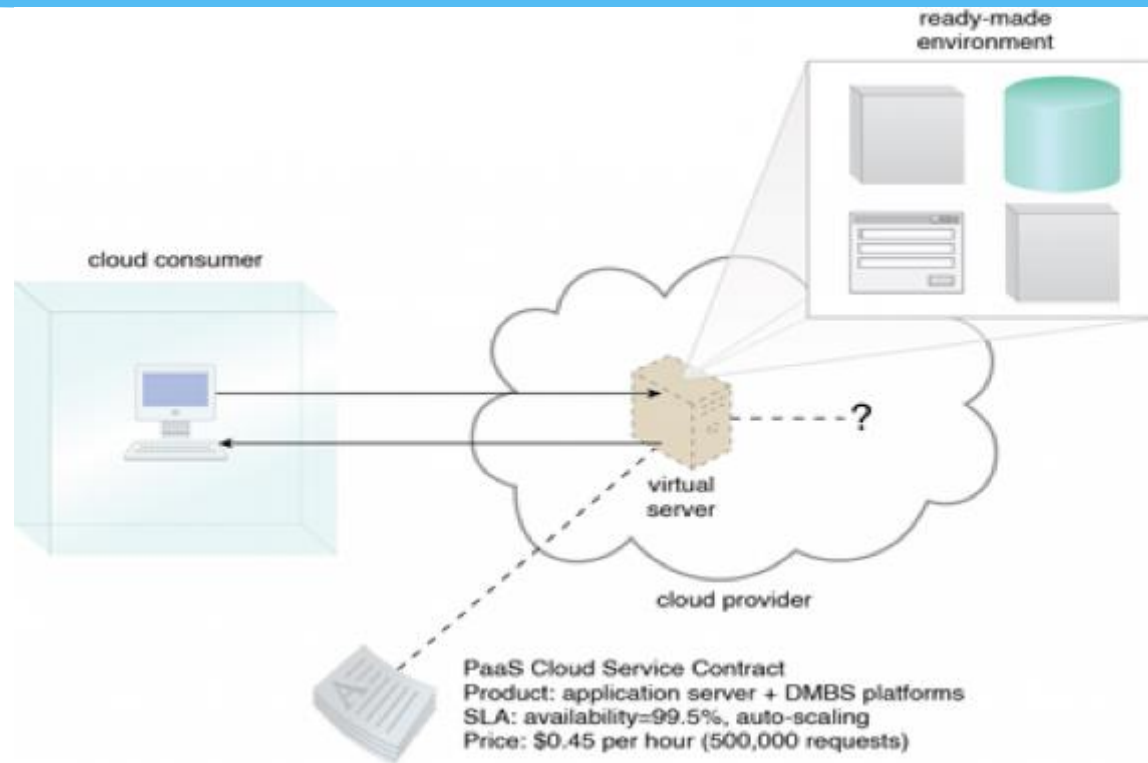
- * Cloud service:
 - * *any IT resource that is made remotely accessible via a cloud*
- * Cloud provider:
 - * *a company or third-party that offers cloud computing services*
- * Cloud consumer:
 - * *an organization or a person that has a formal contract or agreement with a cloud provider to use IT resources that are made available by the cloud provider*
- * Cloud service owner:
 - * *a person or an organization that legally owns a cloud service*
- * Cloud resource administrator:
 - * *a person or an organization responsible for administering cloud-based IT resources (including cloud services)*
- * Cloud carrier:
 - * *provides connectivity between cloud consumers and cloud providers*

Cloud Service Model: IaaS



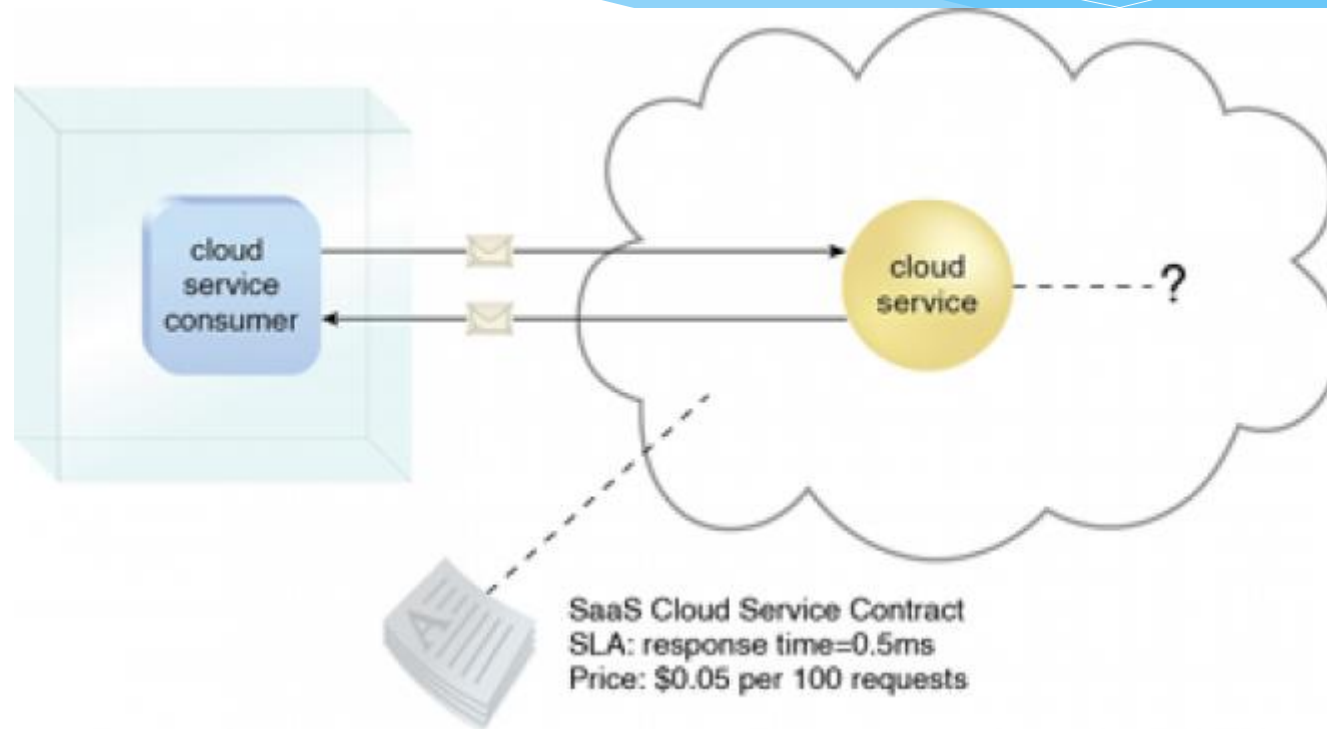
A cloud consumer is using a virtual server within an IaaS environment. Cloud consumers are provided with a range of contractual guarantees by the cloud provider, pertaining to characteristics such as capacity, performance, and availability.

Cloud Service Model: PaaS



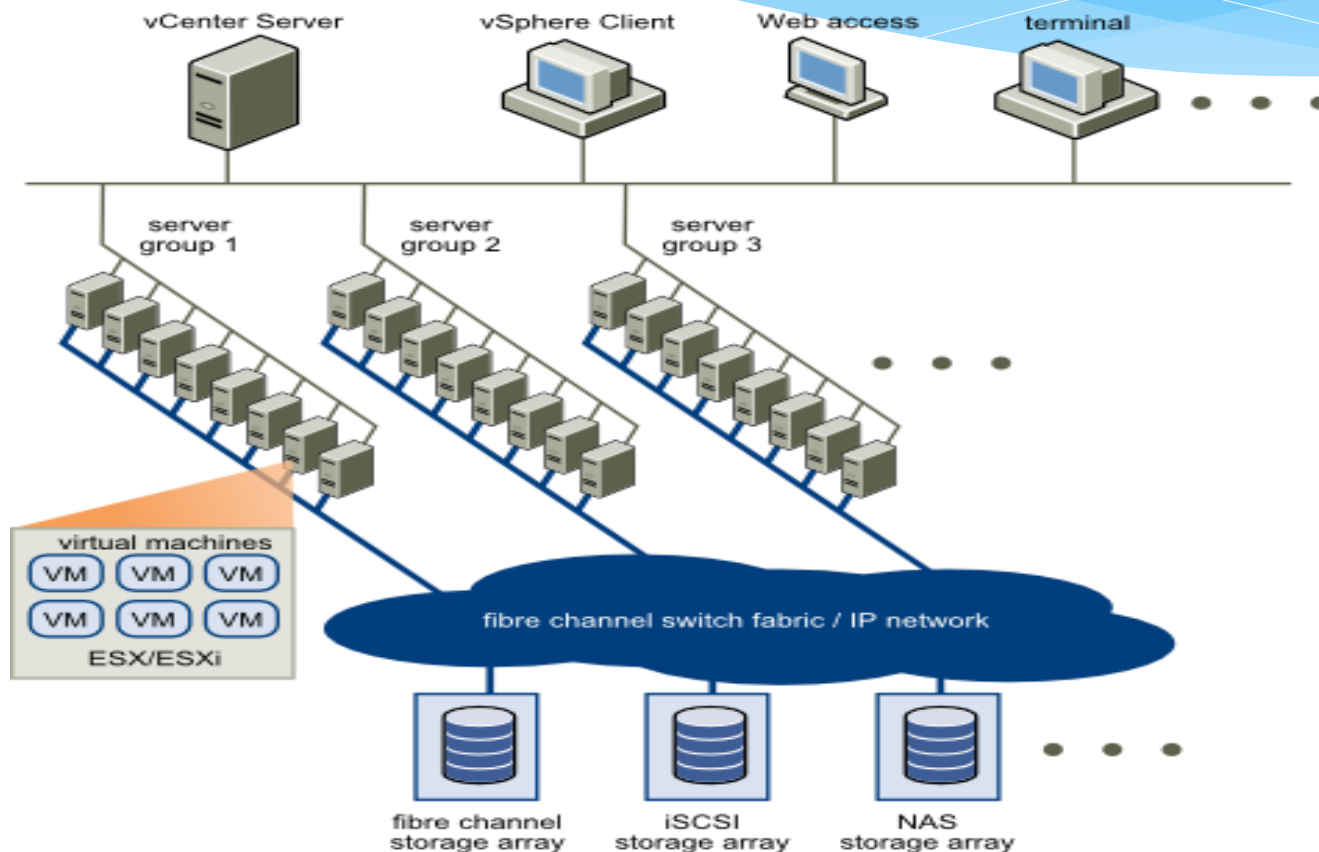
A cloud consumer is accessing a ready-made PaaS environment. The question mark indicates that the cloud consumer is intentionally shielded from the implementation details of the platform.

Cloud Service Model: SaaS



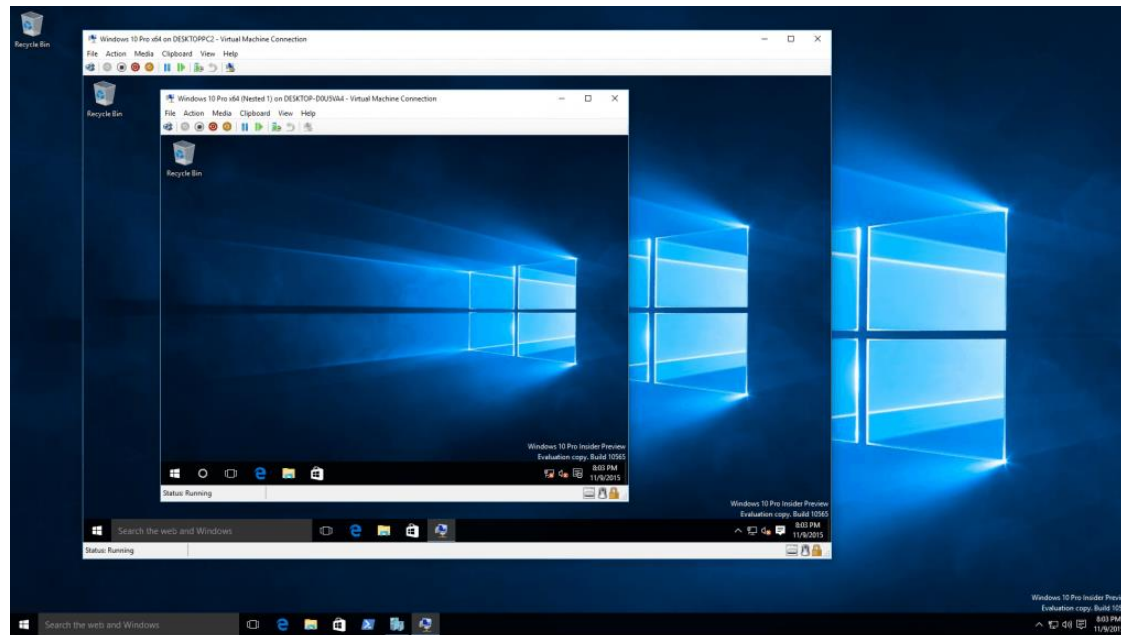
The cloud service consumer is given access the cloud service contract, but not to any underlying IT resources or implementation details.

Physical topology of VMware Infrastructure Data Center



Virtualization

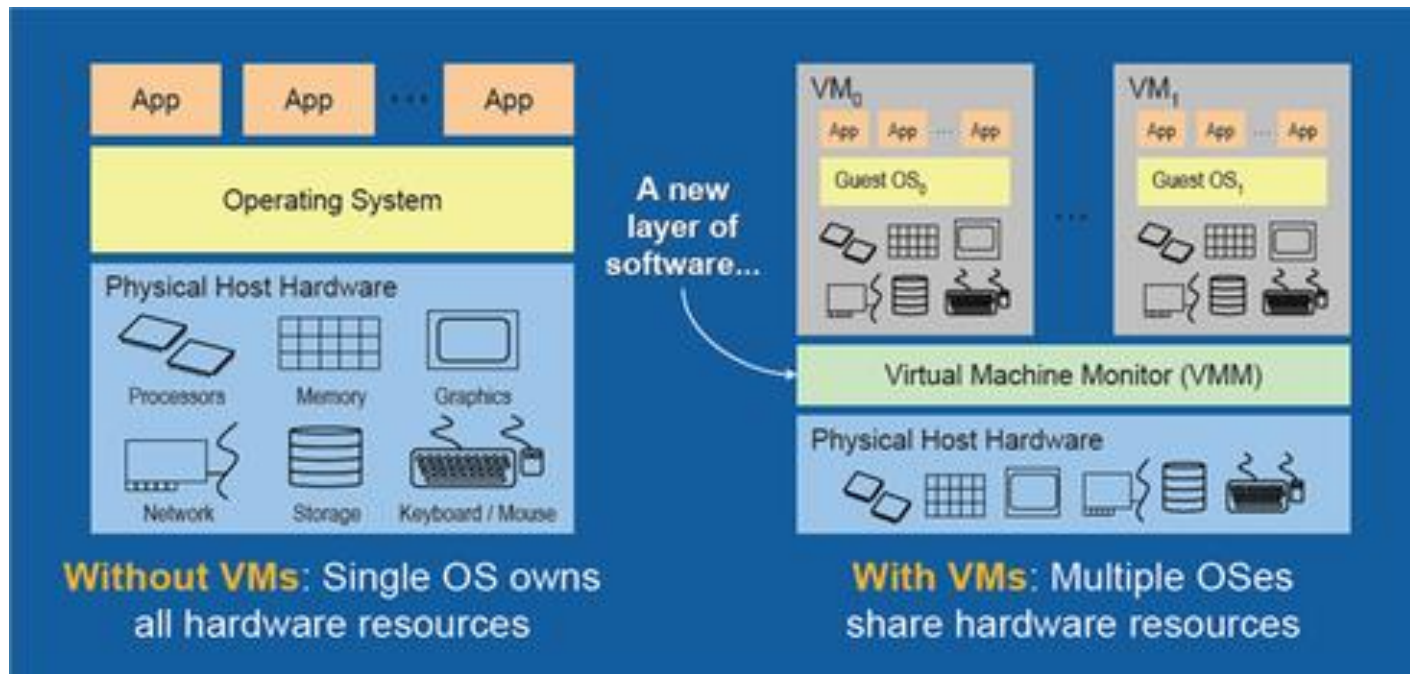
- * Abstraction of computer resources
- * Goal is to collaboratively utilize the IT resources to maximum level and reduce the cost of IT resources



Types of Virtualization

- * Processor virtualization:
 - * enables a processor to be shared across multiple application instances
- * Memory virtualization:
 - * aggregates memory resources into a pool of single memory and manages the memory on behalf of the multiple applications using it
- * Network virtualization:
 - * entails virtual IP management and segmentation
- * Storage virtualization:
 - * provides a layer of abstraction for the physical storage of data

Virtualization



Virtual Machines (Virtual Servers)

- * Virtual machines are isolated instances of the application software and guest operating system that run like a separate computer
- * A VM encapsulates the virtual hardware, virtual disks, and the meta data associated with the application
- * VMs can connect to peripheral devices and function just like physical computers

Hypervisor (Virtual Machine Monitor-VMM)

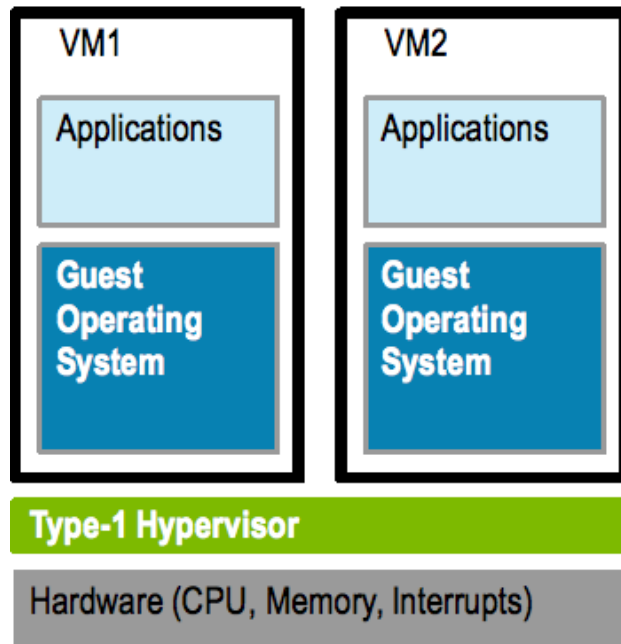
- * The hypervisor is responsible for managing the applications' OSs (guest OSs) and their use of the system resources (e.g., CPU, memory, and storage)
- * It supports isolation and manage multiple VM's running on the same host computer
- * Resources to VM:
 - * Logical CPU and memory
 - * Logical storage blocks
 - * Logical network resources

Functions of Hypervisors

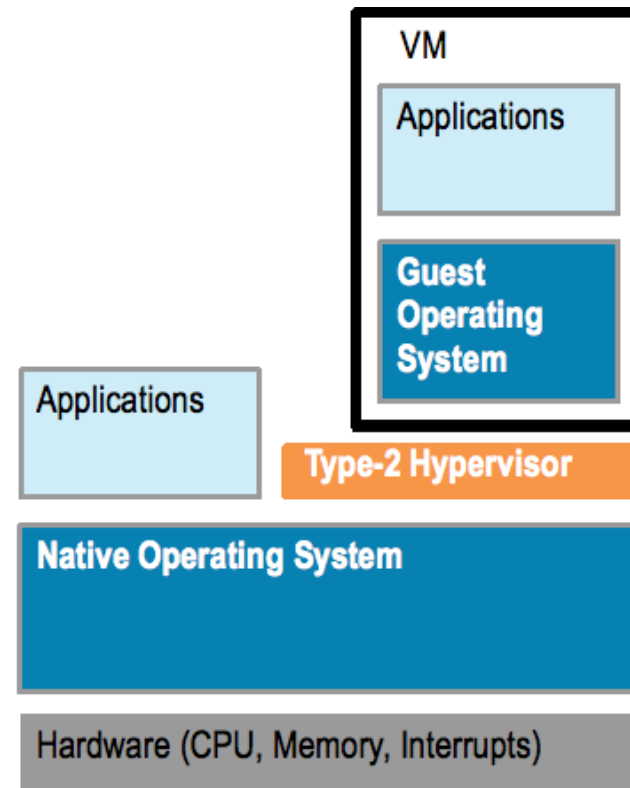
- * Creating and managing VMs
- * Allocating “hardware resources” to VMs from the virtualized pool of hardware resources belonging to the physical server
- * Monitoring the status of the VMs
- * Taking part in the movement of VMs from one system to another

Types of Hypervisors

Bare-metal or Type 1 Hypervisor



Hosted or Type 2 Hypervisor



Typical Cloud Architecture

- * Components

- * Virtual machines



- * Hypervisor



- * Physical Servers



- * Networks



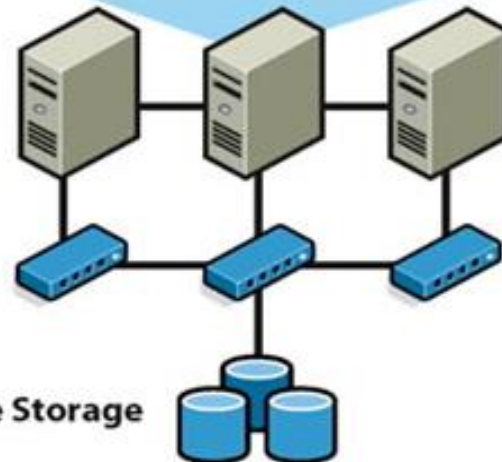
- * SAN Storage



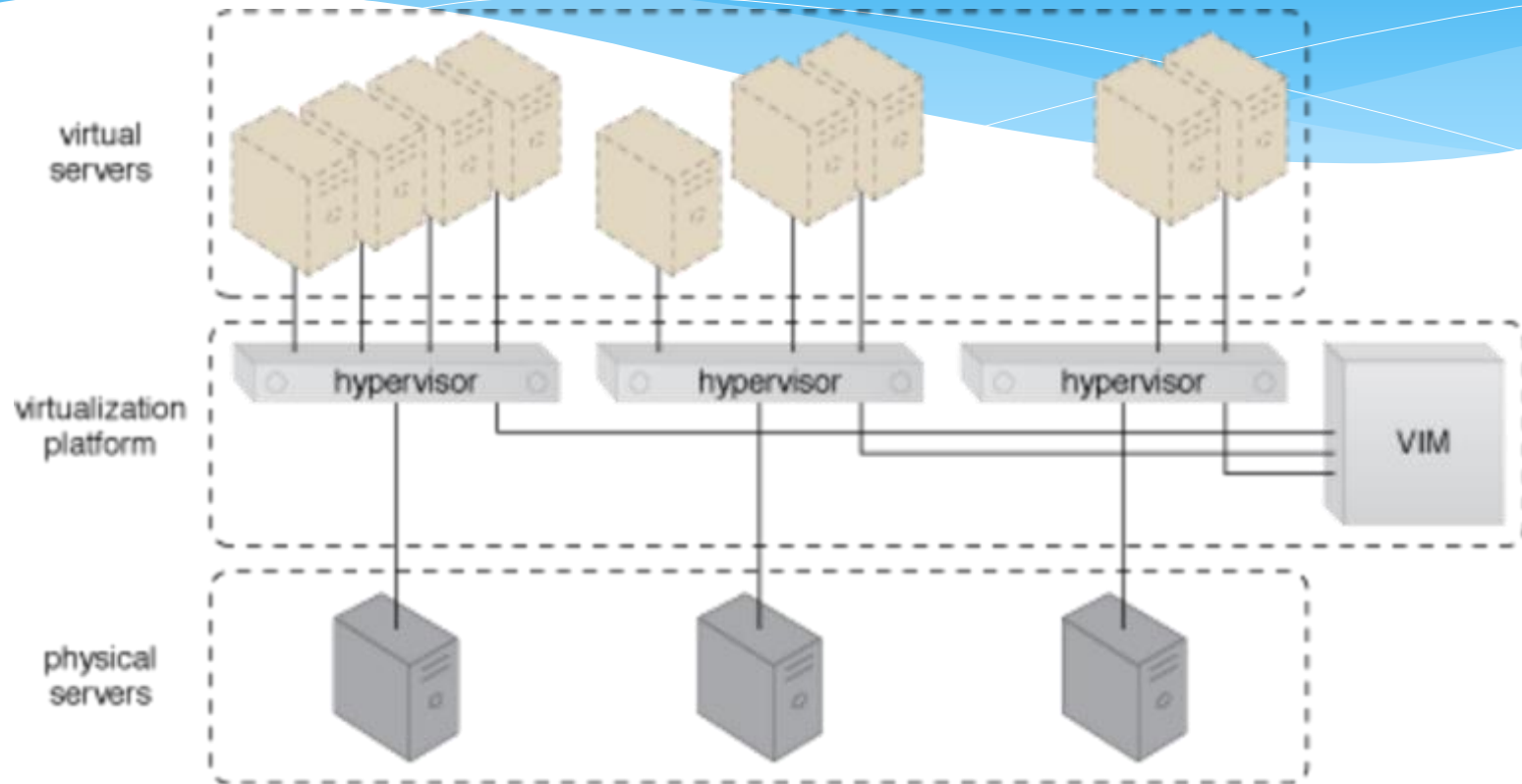
Enterprise Servers

Enterprise Network

Enterprise Storage



Virtual Infrastructure Manager- Hypervisor



Virtual servers are created via individual hypervisors on individual physical servers. All the three hypervisors are jointly controlled by the same VIM (Virtualization infrastructure management).

<http://cloudpatterns.org/mechanisms/hypervisor>

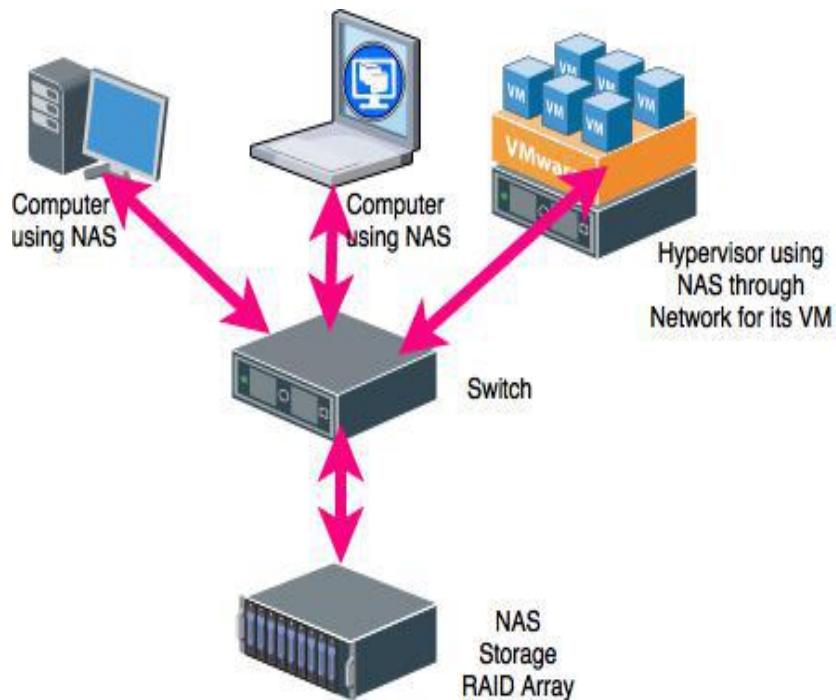
Storage Hardware

- * Direct Attached Storage (DAS)
 - * Storage system directly connected to a computing IT resource (physical server) using a host bus adapter (HBA)

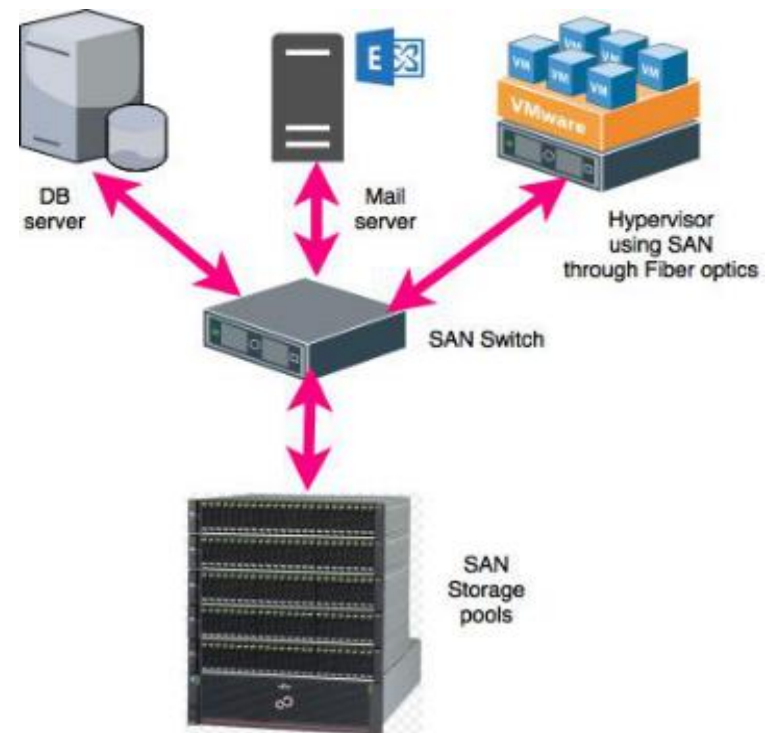


Storage Hardware

* Network-Attached Storage (NAS)



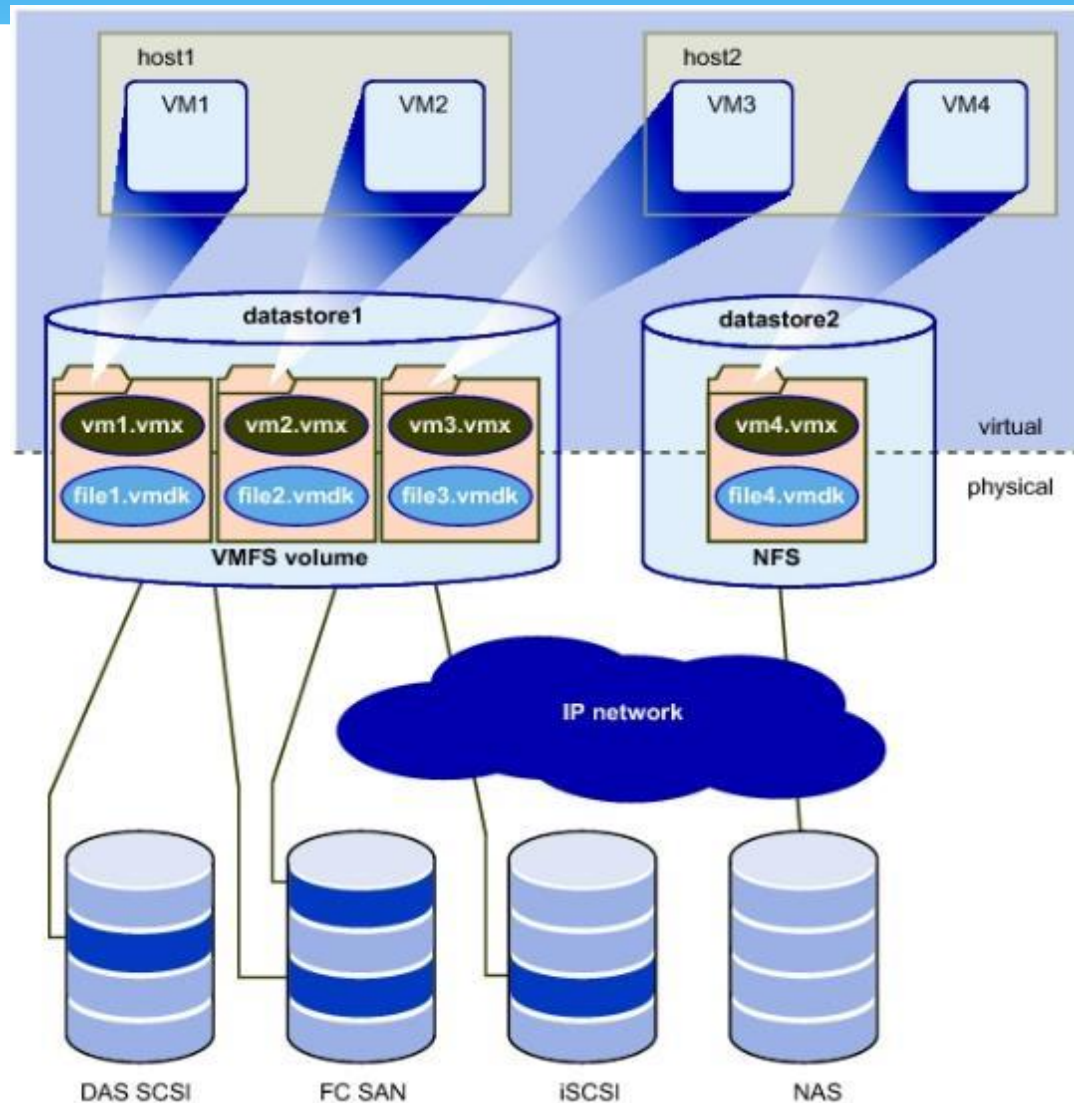
* Storage Area Network (SAN)



Storage Virtualization

- * “the process of presenting a logical view of the physical storage resources to” a host computer system, “treating all storage media (hard disk, optical disk, tape, etc.) in the enterprise as a single pool of storage”
- * Two types of storage technologies:
 - * Block-level virtualization
 - * File-level virtualization
- * Three methods used:
 - * Network-based
 - * Host-based
 - * Array-based

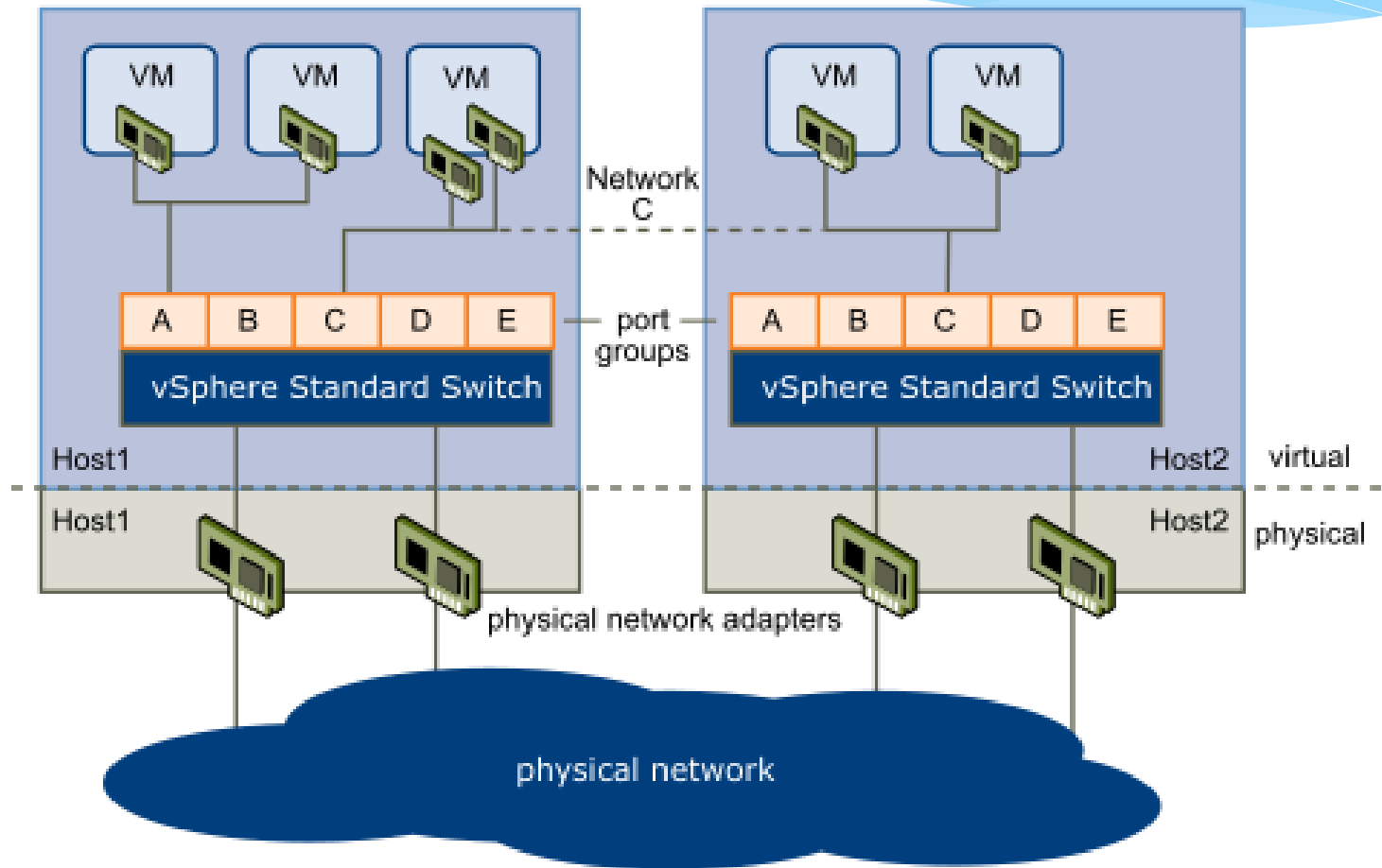
VMware Infrastructure Storage Architecture



Storage Architecture

- * Consists of layers of abstraction that hide and manage the complexity and differences among physical storage subsystems
- * To the applications and guest operating systems inside each VM, storage is presented as SCSI disks connected to a virtual Host Bus Adapter
- * Virtual SCSI disks inside the VMs are provisioned from Datastore elements in the data center

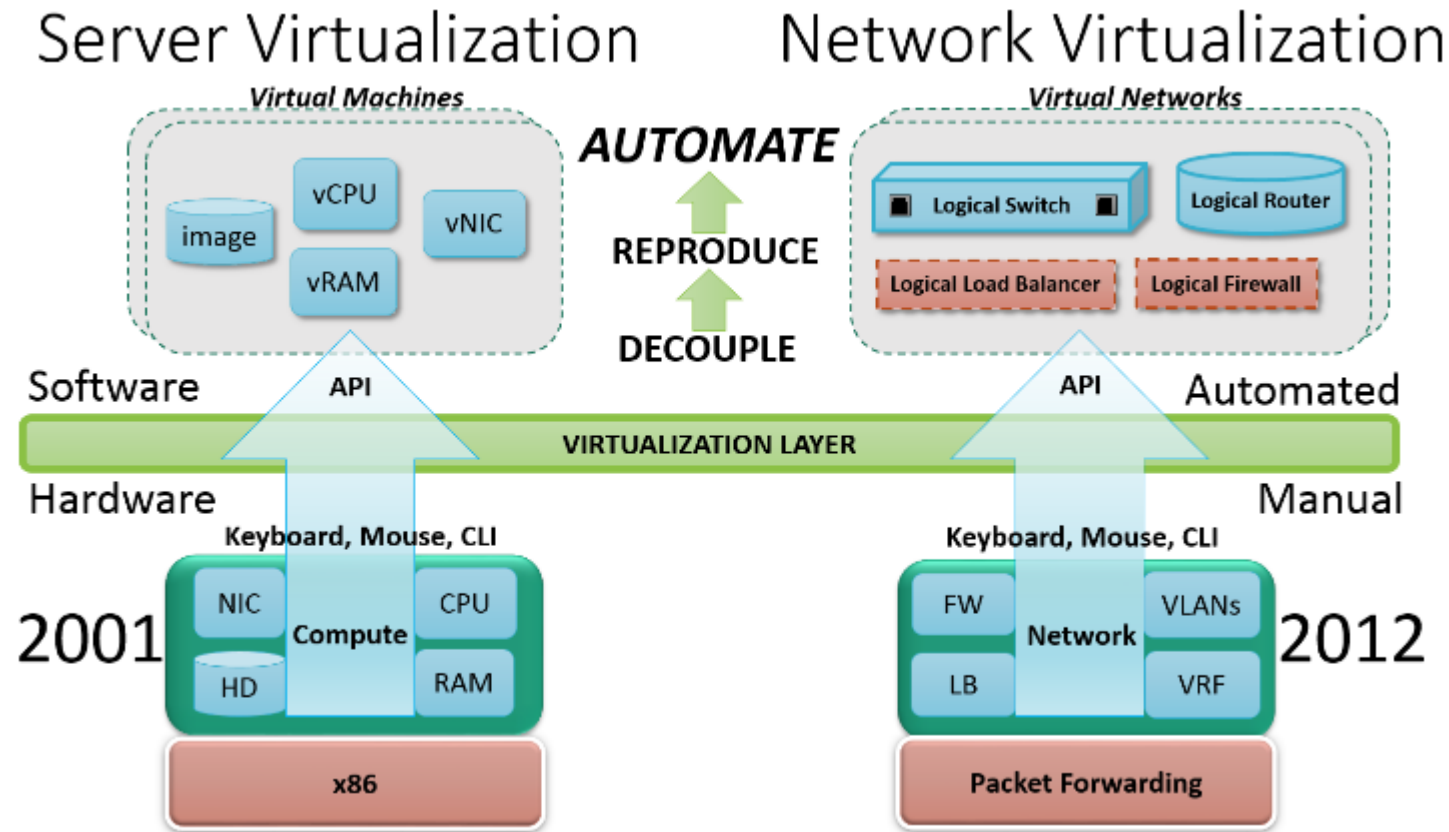
VMware Network Architecture



Network Architecture

- * Virtual machines in a datacentre are like physical machines which are networked in a physical environment
- * Guest OS and application programs communicate with a virtual NIC through either commonly available device driver or a VMware device driver optimized for the virtual environment
- * Port groups aggregate multiple ports under a common configuration and provide a stable anchor point for virtual machines connecting to labelled networks

Network Virtualization



1. <https://www.vmware.com/content/dam/digitalmarketing/vmware/en/pdf/whitepaper/products/nsx/vmware-nsx-network-virtualization-platform-white-paper.pdf>